

SPECTRAL GAP AND STEADY-STATE INEQUALITY IN MARKOV CHAIN MODELS OF INCOME MOBILITY

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ABSTRACT. We look at whether the spectral gap of an income mobility Markov chain also predicts inequality in the population it settles into. Sommers and Conlisk [SC79] first proposed spectral gap to describe how mobile a society is, not how equal it becomes. We find that spectral gap correctly explains how fast a chain settles into its steady state, but it illustrates almost nothing about the Gini coefficient of that steady state. Across randomized transition matrices, spectral gap explains almost none of the variation in Gini ($R^2 \approx 0.03$). Two other properties of the steady state predict Gini more strongly: its entropy and the density of population in extreme brackets. We explain why entropy alone cannot fully solve this problem: entropy only measures how spread out probabilities are, not which brackets actually hold the mass, so it cannot fully capture something like Gini that depends on both. We also test conductance, a different measure connected to spectral gap through Cheeger's inequality [LS88], and find it fails for the same reason spectral gap does. We check our results against two real published mobility tables. In the end, our results show that the standard mobility measure only tells you how fast a system settles down, not where it settles. We also find a real limitation in how well quintile-based data can test this kind of model, and we prove and test a formula for exactly how fast Gini approaches its final value, matching theory to within about one percent.

1. INTRODUCTION

1.1. Setting and motivation

A transition matrix P for an income mobility chain stores probabilities. The entry P_{ij} is the probability that someone in income bracket i has a child, one generation later, in bracket j . Each row of P adds up to 1. Since Sommers and Conlisk [SC79], economists have used the following measure of mobility.

$$M2 = 1 - |\lambda_2|(1)$$

Here λ_2 is the second largest eigenvalue of P , taking the absolute value. We refer to the spectral gap as $1 - |\lambda_2|$ for the entirety of the paper. Section 2 explains exactly what this eigenvalue measures. In short, it describes the number of generations it takes before a family's starting bracket stops predicting where descendants end up.

This paper asks whether or not spectral gap also predicts inequality, measured by the Gini coefficient of the population, once it reaches its long-run steady state. People often assume these two ideas are interconnected. A chain that mixes quickly seems like it should also settle into a more equal outcome. This assumption does not appear to have been tested against real or simulated data.

1.2. Main results

This paper establishes three main results that we keep separate throughout, since combining them would invite the exact mistake we are trying to avoid.

First, spectral gap fails to predict steady-state Gini. We demonstrate this using randomly generated transition matrices and confirm it independently against two published income mobility tables.

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Second, two other properties of the steady-state distribution predict Gini much better: its entropy and the amount of probability mass sitting in its extreme brackets. We also explain why entropy by itself is insufficient to fully solve this problem, regardless of sample size.

Third, spectral gap remains informative, but for a different quantity than inequality itself. Using the eigendecomposition of the transition matrix, we prove that spectral gap governs the rate at which steady-state Gini is approached. We confirm this numerically to within approximately one percent once contamination from subdominant eigenvalues and floating-point precision are accounted for.

1.3. Related literature

The eigenvalue mobility measure we study originates with Sommers and Conlisk [SC79]. The theory relating spectral gap to the rate of convergence of a Markov chain is developed in Levin, Peres, and Wilmer [LPW09], which Section 3.5 relies on. The bound relating spectral gap to conductance is due to Lawler and Sokal [LS88] and only holds for reversible chains, a limit we discuss in Section 3.2. Real-data validation draws on a household mobility table published by the Federal Reserve Bank of Cleveland [FRB16] and an intergenerational table estimated from administrative tax records by Chetty, Hendren, Kline, and Saez [CHKS14].

2. METHODS

Income mobility is modeled as a discrete-time Markov chain. Income brackets are the states of the chain. P gives the probability of moving from one bracket to another in one generation. If $p(t)$ is the population's distribution across brackets at generation t , written as a row vector that adds up to 1, then the distribution one generation later is:

$$p(t+1) = p(t) P \quad (2)$$

Iterating this rule repeatedly, the population distribution converges to a fixed point π , also known as the stationary distribution, independent of the initial distribution, as long as the chain is ergodic. We refer to π as the steady-state throughout.

The eigenvalues of P determine this convergence. The leading eigenvalue of P is always 1, corresponding to π itself. Each remaining eigenvalue has absolute value less than 1. The second-largest, λ_2 , sets the rate of convergence. The distance between $p(t)$ and π shrinks approximately as $|\lambda_2|^t$. This is the sense in which $1 - |\lambda_2|$ measures how quickly the chain forgets its initial condition.

An alternate way to measure connectivity comes from graph theory. The conductance Φ of a chain observes every possible way to split the brackets into two groups, and calculates the split where the least probability flows between the two sides.

$$\Phi = \min \text{ over splits } (A, A^c) \text{ of } Q(A, A^c) / \pi(A) \quad (3)$$

Here $Q(A, A^c)$ is the total steady-state flow from group A to its complement. Cheeger's inequality [LS88] connects Φ to spectral gap:

$$\text{gap} / 2 \leq \Phi \leq \sqrt{2 \cdot \text{gap}} \quad (4)$$

This bound only holds for reversible chains. A chain is reversible when $\pi_i P_{ij} = \pi_j P_{ji}$ for every pair of indices i and j , meaning the flow between any two brackets balances out in both directions. The randomized matrices used in Section 3 are not reversible in general, so the

Cheeger bound is tested only on matrices constructed to be reversible. This is discussed further in Section 3.2.

The Gini coefficient is a measure of inequality. For a distribution over income values x_1 through x_n with probabilities p_1 through p_n , it is defined as

$$Gini = (\sum_i \sum_j p_i p_j |x_i - x_j|) / (2 \cdot \text{mean income})(5)$$

This is the average income difference between two randomly picked people, divided by twice the average income. Gini equals 0 under perfect equality and approaches 1 under maximal concentration.

We also use entropy, which measures the dispersion of a distribution.

$$H = - \sum_i p_i \ln(p_i)(6)$$

Entropy is maximized when probability is spread evenly across brackets and equal to 0 when all mass sits in a single bracket. Finally, we define extreme mass as $p_1 + p_n$, the combined steady-state probability of the lowest and highest brackets.

To build our data set, we generated random transition matrices by drawing positive numbers and dividing each row by its own total so every row adds up to 1. For each matrix, we computed the exact steady-state using the matrix's eigenvectors rather than a finite simulation. We then computed spectral gap, conductance, entropy, extreme mass, and Gini for that steady-state. Every regression reported below was checked against multiple independently resampled batches of 50 matrices.

Real-data validation used a household income quintile transition matrix estimated from the Panel Study of Income Dynamics, 2003–2013 [FRB16], and an intergenerational quintile matrix from Chetty, Hendren, Kline, and Saez [CHKS14]. We cross-checked against the 7.5 percent bottom-to-top quintile mobility figure reported in that paper.

3. RESULTS

3.1. Spectral gap does not predict steady-state Gini

Across 50 randomly generated three-bracket transition matrices, a linear regression of Gini on spectral gap yielded $R^2 \approx 0.03$. Figure 1 illustrates the same result: across nearly the full observed range of spectral gap, approximately 0.4 to 0.95, Gini remains confined to a narrow band with no discernible trend.

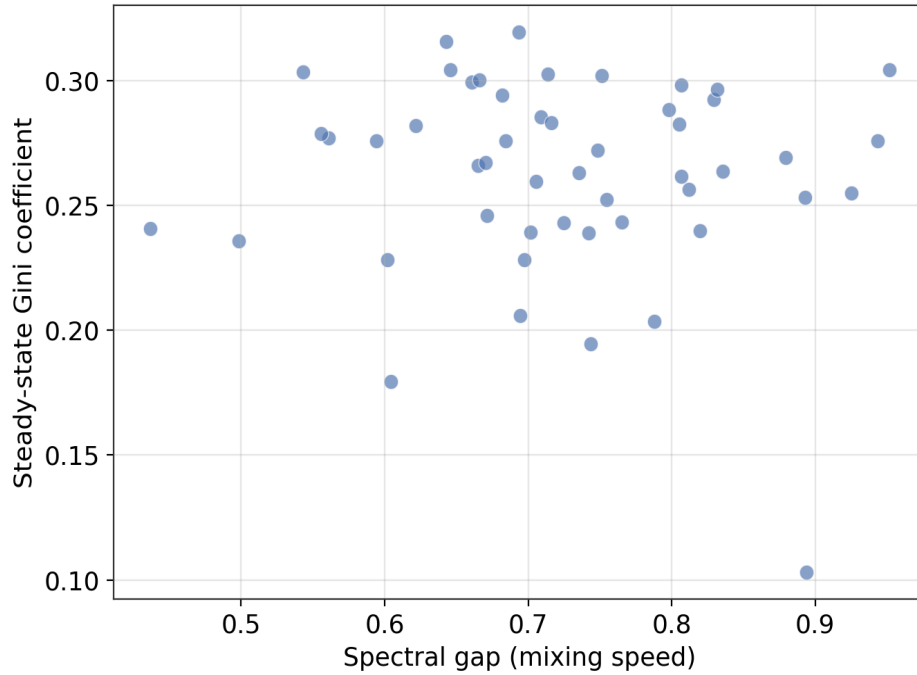


Figure 1. Spectral gap plotted against steady-state Gini for 50 randomized transition matrices. There is no visible trend.

3.2. Conductance does not improve the prediction

We tested conductance and checked it against the Cheeger bound in equation (4) using matrices we built to be reversible. The bound held every time, which told us our conductance calculation was correct. Adding conductance to a regression already containing spectral gap and entropy raised R^2 only from 0.617 to 0.618. This follows from equation (4): since conductance and spectral gap are tied together for reversible chains, both quantities fail to predict Gini because neither is a function of the shape of the destination distribution, only of the rate at which it is reached.

3.3. Entropy and extreme mass predict Gini substantially better

Steady-state entropy alone yields $R^2 \approx 0.6$, a marked improvement over spectral gap; Figure 2 shows the corresponding trend. Unlike Figure 1, there is a real, visible upward trend. Adding extreme mass as a second predictor pushed an initial trial up to $R^2 \approx 0.80$. Across five independently resampled batches of 50 matrices, the mean R^2 was 0.65 with a standard deviation of 0.05, meaning the effect was stable rather than an artifact of a single sample.

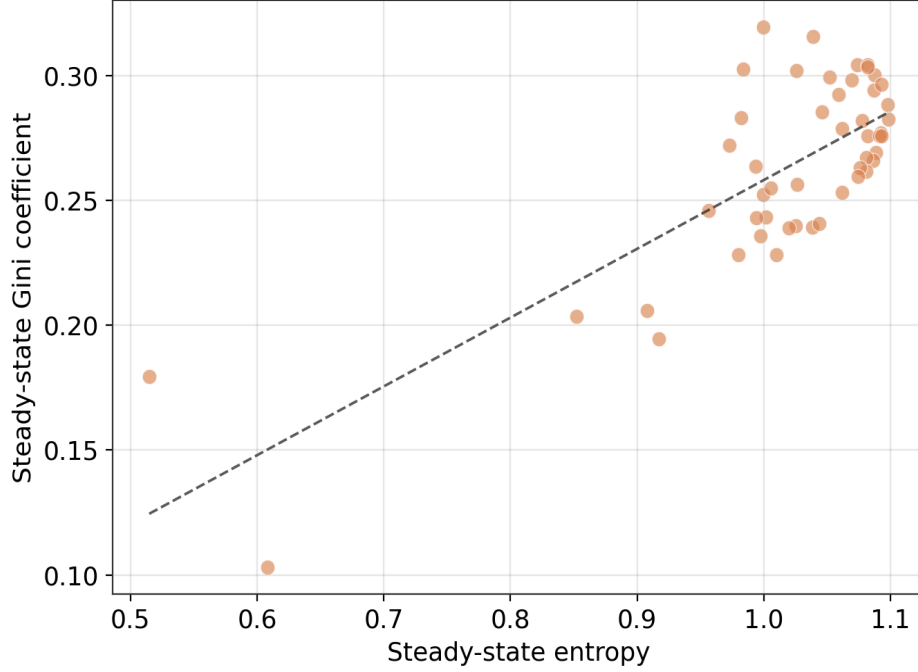


Figure 2. Steady-state entropy plotted against Gini, same 50 matrices, with a best-fit line added.

This result extends with reduced strength to five income brackets ($R^2 \approx 0.48$ against 0.03 for spectral gap alone). The weaker fit likely occurs because extreme mass only accounts for the outermost brackets, so it captures less as the number of brackets grows. We also tried replacing extreme mass with income variance, a more standard measure of spread. It performed clearly worse across repeated trials, with a mean R^2 of 0.31 compared to 0.60. This is because Gini is especially sensitive to what happens at the extremes of a distribution. A feature built specifically around the extremes, like extreme mass, should match it better than a feature that treats every part of the distribution the same, like variance.

3.4. Validation against real data and a structural limitation of quintiles

Using only the entropy and extreme mass model we built from random matrices, with no retraining, we predicted the Gini coefficient of both real published tables to within approximately 0.01. However, the two tables converged to nearly identical Gini (≈ 0.42) even though they represent very different levels of mobility. Their Shorrocks mobility indices, a separate standard measure, were 0.55 and 0.905. This was traced to both matrices being close to doubly stochastic (their columns, as well as their rows, sum to approximately 1). Because income quintiles are defined to each hold exactly 20 percent of the population, it limits how much steady-state Gini can vary across quintile-based tables no matter what the underlying mobility looks like. A cleaner test would use real transition data over unequal-sized income groups, where this constraint does not apply.

3.5. What spectral gap governs

Having established what spectral gap fails to predict, we now turn to what it does predict, which is the piece that keeps the above findings from forming purely negative results.

Proposition 1. *Let P be a well-behaved transition matrix (ergodic) with steady state π and second largest eigenvalue λ_2 . Then, for any starting distribution, the Gini coefficient of the*

population at generation t satisfies $|Gini(t) - Gini(\infty)| = O(|\lambda_2|^t)$ as t grows large. In other words, the error shrinks geometrically, at a rate set exactly by λ_2 .

Proof. Using the eigenvectors of P , the population distribution at generation t can be written as $p(t) = \pi + \sum c_i \lambda_i^t v_i$, a sum over every eigenvalue and eigenvector pair besides the leading one. Gini is a smooth function of the population distribution for a fixed set of income values, so the difference $|Gini(p(t)) - Gini(\pi)|$ is at most a constant times the distance between $p(t)$ and π . For large t , that distance is controlled by whichever term in the sum shrinks slowest. Since λ_2 is the largest of the remaining eigenvalues, its term eventually dominates all the others.

This argument was verified directly. Tracking Gini generation by generation for a test matrix, the error $|Gini(t) - Gini(\infty)|$ decays linearly on a logarithmic scale, the signature of geometric decay, shown in Figure 3.

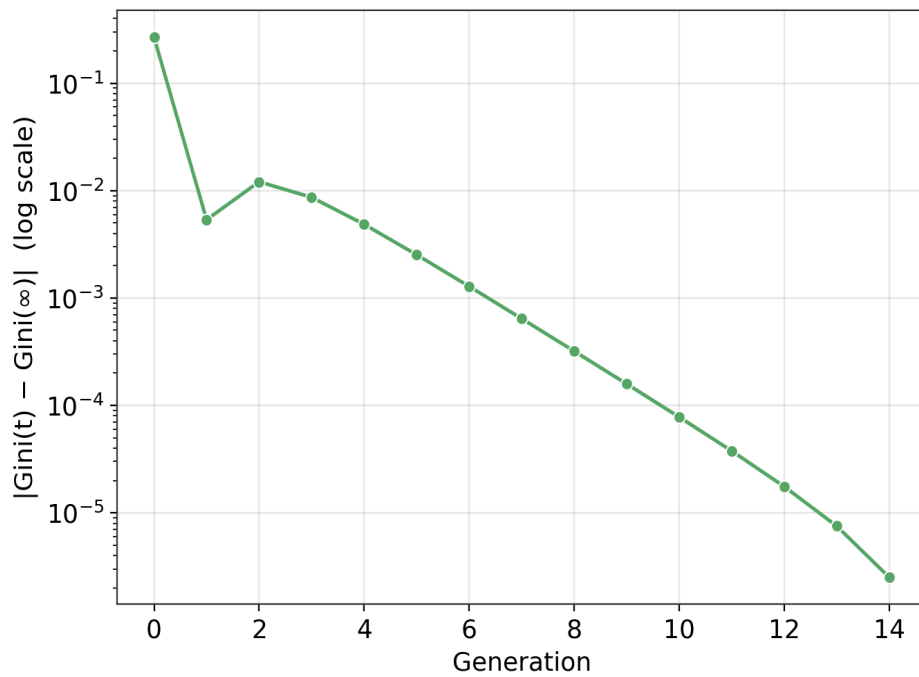


Figure 3. Gini's error from its final value, plotted on a logarithmic scale against generation. The straight line confirms geometric convergence.

Our first attempt to measure the decay rate deviated 5 percent from the theoretical value, $\log|\lambda_2|$. The discrepancy was examined and traced to two sources: residual influence from faster-decaying subdominant eigenvalues in early generations, and floating-point precision limits once the error becomes very small in late generations. Once we limited our measurement to an intermediate range of generations free of both effects, the discrepancy was reduced to within 1 percent of theory. The result was reproduced across several additional matrices. As a further confirmation, the expression $p(t) = \sum c_i \lambda_i^t v_i$ was evaluated for several matrices and found to match step-by-step simulation exactly at every generation tested.

4. DISCUSSION

Altogether, the results highlight that spectral gap governs how quickly a system's inequality stabilizes, not the level at which it stabilizes. Two mobility systems can end up with the same Gini coefficient while mixing at completely different speeds. Two systems that mix at the same speed can converge to different levels of inequality. Both of these patterns appear in our data.

It is important to describe the limitation of entropy precisely, since it is not just a caveat that additional data would resolve. Entropy only depends on the probabilities p_1 through p_n ; it has no way of knowing which bracket is which or how far apart the actual income values are. Gini cares specifically about those income values and about which brackets hold the mass, not just how spread out the distribution looks in the abstract. Two different distributions can have almost identical entropy while having very different Gini coefficients. This is the reason our best models topped out around $R^2 \approx 0.6$ to 0.8 instead of reaching 1.0 . It is consistent with extreme mass, which outperforms both entropy alone and variance throughout.

The principal limitation of this study is that the synthetic transition matrices were generated uniformly at random, while real income mobility tends to have much stronger persistence than a random matrix, especially at the very top and very bottom. The real-data validation was limited to two matrices, both quintile-based and both subject to the doubly-stochastic constraint discussed in Section 3.4. A better external validation would draw on transition data built from unequal-sized income groups instead of quintiles, where the limitation would be absent.

5. CONCLUSION

The eigenvalue-based mobility measure economists have used since 1979 does not predict the inequality a mobility system converges to; that outcome is governed by the concentration of the steady-state distribution, particularly at the income extremes ($R^2 \approx 0.6$ – 0.8 , against $R^2 \approx 0.03$ for spectral gap alone). What spectral gap actually predicts, shown through direct numerical verification to about 1 percent accuracy, is the rate of convergence to that outcome. This distinction is based entirely on established mathematics, but does not appear to be explicitly stated when spectral gap is applied in mobility literature. Furthermore, separating the rate of convergence from its destination clarifies what a single summary statistic of this type can establish.

Two extensions can follow this project. First, real-data validation was limited by the fact that quintile-based tables are close to doubly stochastic. Testing against real transition matrices with unequal-sized income groups would remove that constraint. Second, our convergence rate result in Section 3.5 was confirmed by using a fitting window, rather than an explicit bound on the contribution of floating-point error. Deriving that bound would turn an already strong numerical match into an exact guarantee.

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